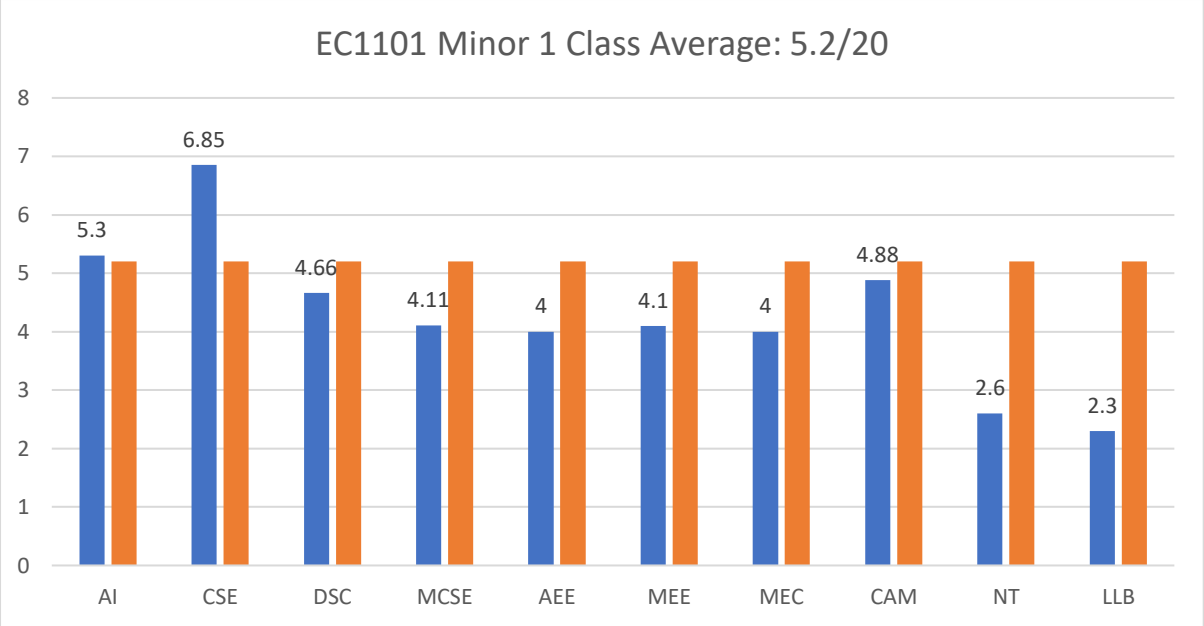
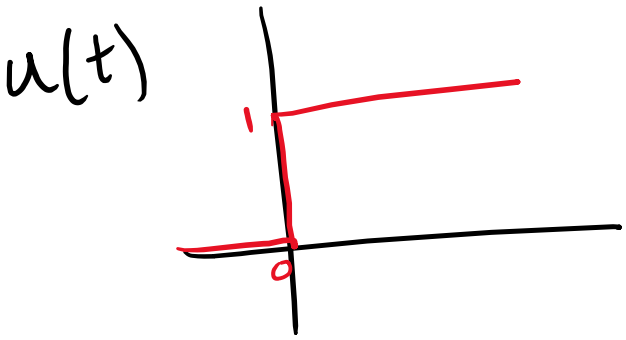
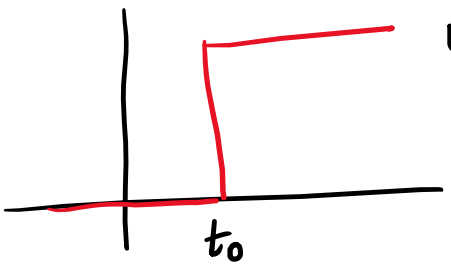

EC1101 Minor 1 Class Average: 5.2/20
8
7
6
5
4
3
2
1
0
5.3
6.85
4.66
4.11
4
4.1
4
4.88
2.6
2.3
AI
CSE
DSC
MCSE
AEE
MEE
MEC
CAM
NT
LLB

Step Response of RC/RL circuits.

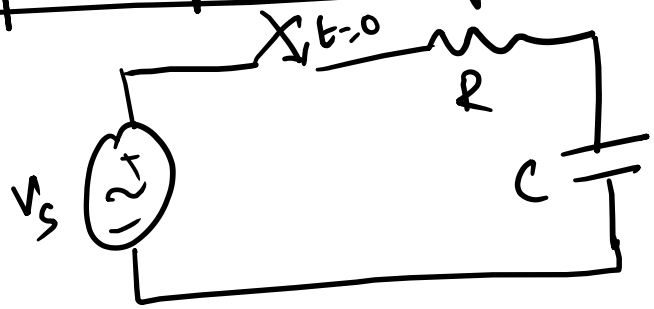
$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$

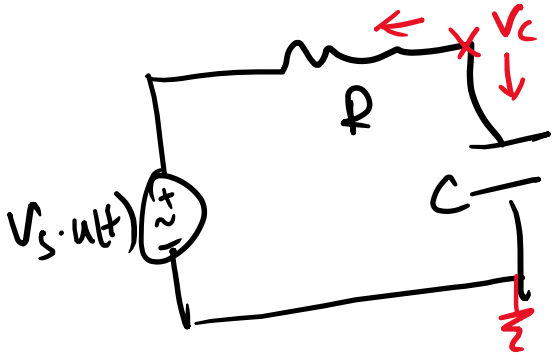




$$u(t-t_0) = \begin{cases} 0 & t < t_0 \\ 1 & t \geq t_0 \end{cases}$$

Step Response of RC





$$\frac{V_c - V_s \cdot u(t)}{R} + C\frac{dV_c}{dt} = 0$$

$$\frac{dV_c}{dt} + \frac{V_c}{RC} = \frac{V_s \cdot u(t)}{RC}$$

For $t > 0$ $\quad \frac{dV_c}{dt} + \frac{V_c}{RC} = \frac{V_s}{RC}$

$$\frac{dV_c}{dt} = -\frac{(V_c - V_s)}{RC} \Rightarrow \frac{dV_c}{V_c - V_s} = \frac{-dt}{RC}$$

Integrate on both sides.

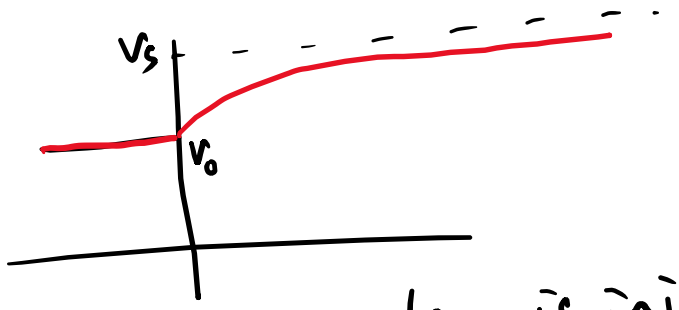
$$\ln(v_c - v_s)\Big|_{v_0}^{v_c(t)} = -\frac{t}{RC}\Big|_0^t$$

$$\ln\frac{v_c(t)-v_s}{v_0-v_s} = \frac{-t}{RC}$$

$$\boxed{\tau = RC}$$

$$v_c(t) = v_s + (v_0 - v_s)\,e^{-t/\tau}$$

$$v_c(t) = \begin{cases} v_0 & t < 0 \\ v_s + (v_0 - v_s)e^{-t/\tau} & t \geq 0 \end{cases}$$



Assume capacitor is initially not charged.

$$v_c(t) = \begin{cases} 0 & t < 0 \\ v_s(1 - e^{-t/\tau}) & t \geq 0 \end{cases}$$

$$v_c(t) = v_s(1 - e^{-t/\tau})\,u(t)$$

Complete Response = natural response + forced response.

$$v = v_n + v_f$$

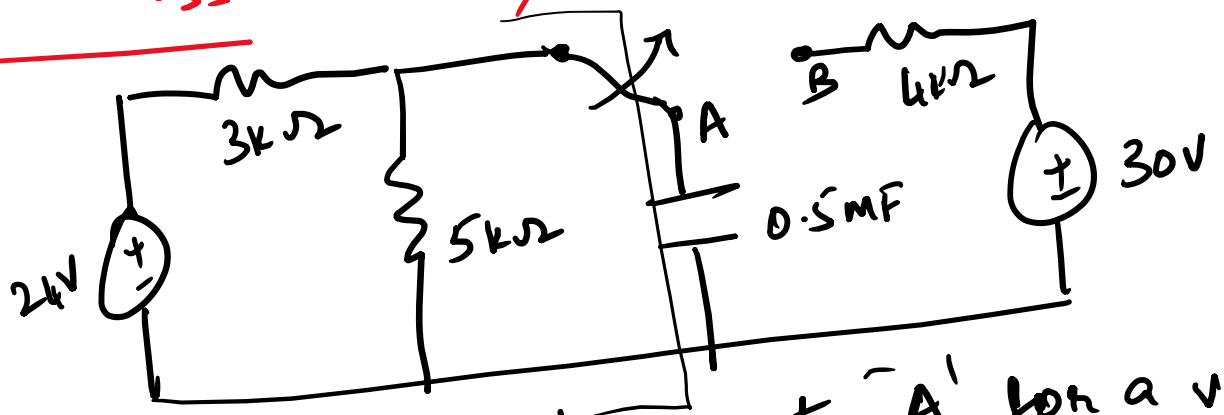
$$v_n = v_0 e^{-t/\tau}$$

$$v_f = v_s(1 - e^{-t/\tau})$$

Complete Response: transient + steady state.

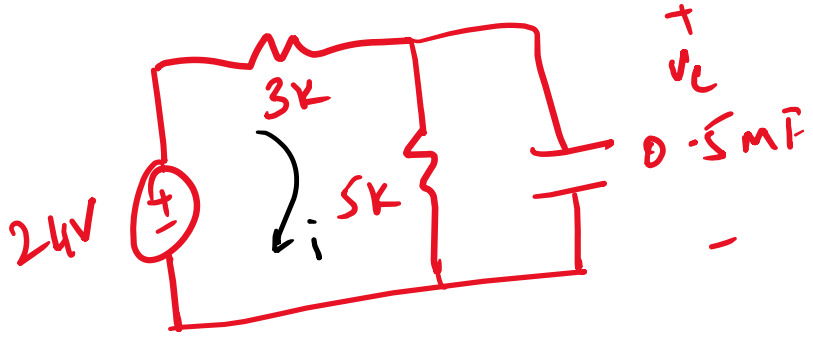
$$v = v_t + v_{ss}$$

$$v_{ss} : v_s \quad ; \quad v_t = (v_0 - v_s)\, e^{-t/\tau}$$



- Switch has been at 'A' for a very long time.
- At $t=0$, switch moves to 'B'. Find $v_c(t)$ for $t \geq 0$

For $t < 0$



$$i = \frac{24}{3K+5K}$$

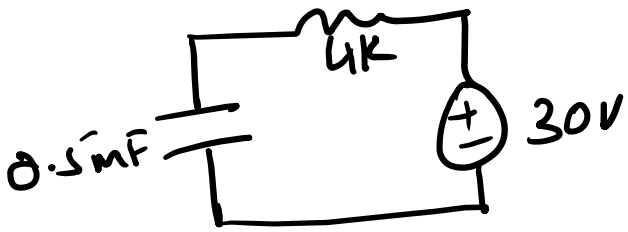
$$v_C = v_{5K} = \frac{24 \cdot 5K}{8K}$$

$$v_c = 15V$$

$$v_C(0) = 15V$$

$$v_C(0^-) = v_C(0) = v_C(0^+) = 15V$$

For $t \geq 0$



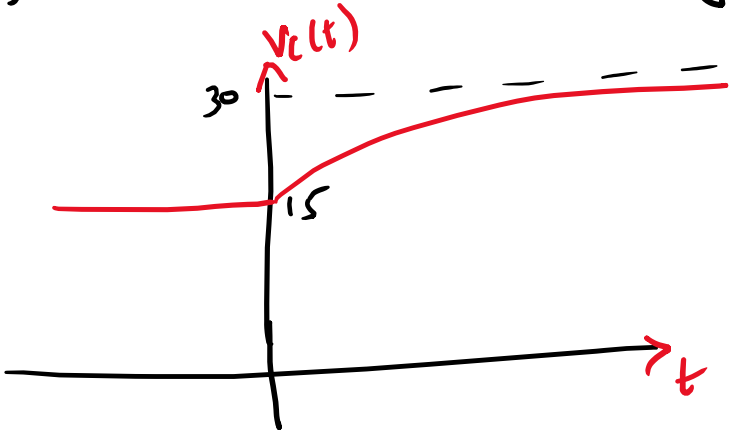
$v(0) = 15V$

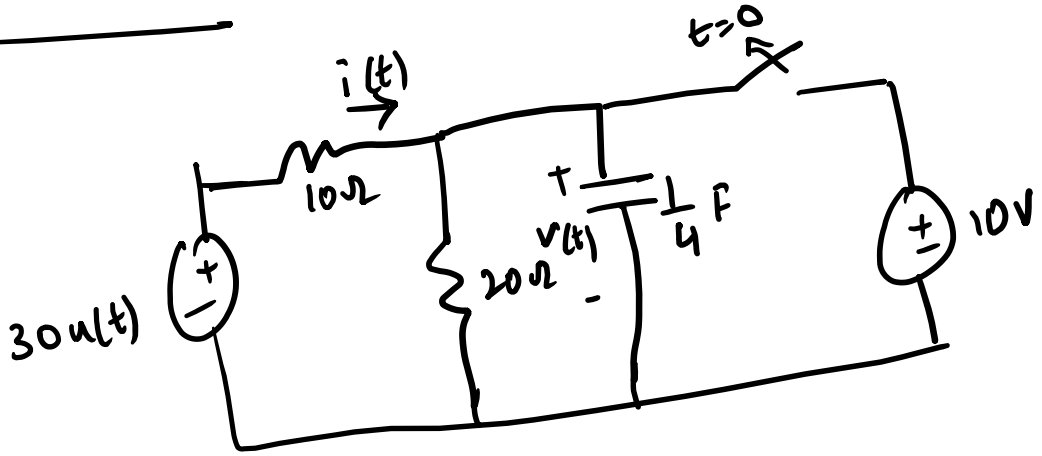
$v(\infty) = 30V$

$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$

$= 30 + [15 - 30]e^{-0.5t}$

$v(t) = 30 - 15e^{-0.5t}$ V for $t \geq 0$

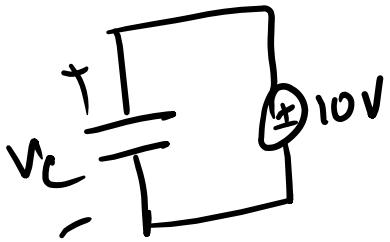




Switch opens at $t = 0$.

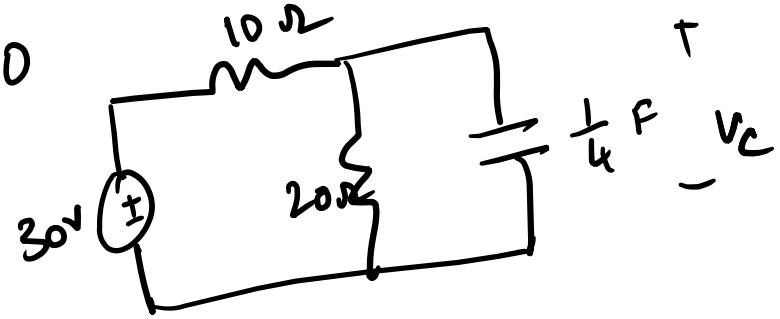
Find $v(t)$, $i(t)$

For $t < 0$



$v_c = 10V$

For $t > 0$



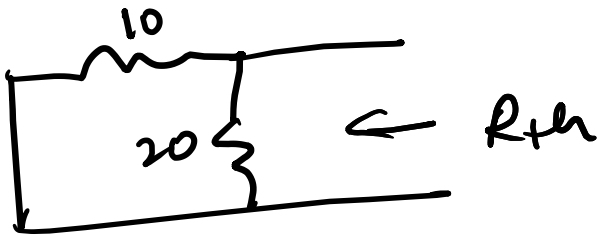
$v_c(0) = 10V$

$v_c(\infty) = 20V$

$\tau = RC$

$R = R_{th}$

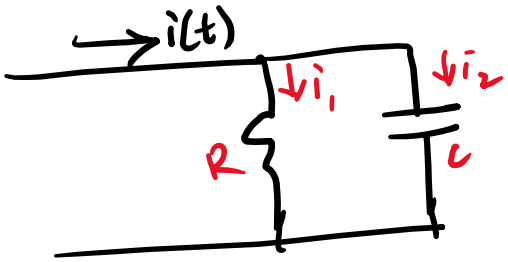
$R_{th} = \frac{20}{3}\,\Omega$



$\tau = R_{th} \cdot C = \frac{20}{3} \cdot \frac{1}{4} = \frac{5}{3}\,s$

$$v_c(t) = v(\infty) + [v(0) - v(\infty)]\, e^{-t/\tau}$$

$$v_c(t) = 20 - 10e^{-0.6t} \quad \text{for } t \geq 0$$



$i(t) = i_1 + i_2$

$$i(t) = \underbrace{\frac{v_c(t)}{R}}_{i_1} + \underbrace{C\frac{dv_c(t)}{dt}}_{i_2}$$